Title (Units): **COMP7015 Artificial Intelligence (3,2,1)**

Course Aims: To provide students with a comprehensive understanding of the fundamental principles and practical applications of modern Artificial Intelligence, with a strong emphasis on deep learning methodologies and their use in addressing contemporary challenges across various domains.

Prerequisite: Nil

Course Intended Learning Outcomes (CILOs):
Upon successful completion of this course, students should be able to:

No.	Course Intended Learning Outcomes (CILOs)
	Knowledge
1	Explain the capabilities, strengths and limitations of various deep learning algorithms
2	Describe the fundamental principles and building blocks of deep learning, including neural networks, loss functions, and optimization techniques
3	Explain the characteristics, training and fine-tuning methodologies, and ethical considerations surrounding foundation models, large language models, and multimodal generative AI
	Professional Skill
4	Determine the suitable deep learning algorithms for various real-world problems
5	Train and evaluate deep learning algorithms to solve real-world problems
6	Apply post-training and retrieval augmented generation to utilize pre-trained foundation models

Calendar Description: This course provides a comprehensive introduction to the principles and practical applications of modern Artificial Intelligence, with a strong focus on deep learning. Students will explore fundamental deep learning concepts, architectures such as convolutional and recurrent neural networks, attention mechanisms, and transformer models. The course will also cover the emerging field of foundation and large language models, as well as multimodal deep learning, equipping students with insights into the cutting-edge advancements in AI.

Teaching and Learning Activities (TLAs):

CILOs	Type of TLA
1-3	Students will learn the AI principles and techniques in lectures.
4-6	Students will acquire hands-on experience via practical assignments and course project.

Assessment:

No.	Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
1	Continuous Assessment	50%	1-5	Written assignments and quiz will be used to evaluate students' understanding of theoretical concepts of AI. Practical assignments and hands-on course project will be used to assess students' ability to apply the deep learning algorithms to solve real world problems.
2	Examination	50%	1-5	Examination will be used to assess students' overall understanding of various AI algorithms, their applications, as well as their capabilities, strengths, and limitations.

Assessment Rubrics:

Excellent (A)	• Achieve the six CILOs, demonstrating an excellent mastery of both the theoretical and practical aspects of the knowledge and skills associated with AI

	• Able to develop correct solutions to real-world problems • Demonstrate a thorough understanding and solid knowledge of AI • Able to apply a variety of techniques and relevant knowledge for solving problems in AI
Good (B)	• Achieve most of the six CILOs, demonstrating a good understanding of the knowledge and skills associated with AI • Able to develop correct solutions to real-world problems • Demonstrate a competent level of knowledge of AI • Ability to make use of appropriate techniques and knowledge and apply them to familiar problems
Satisfactory (C)	• Achieve some of the six CILOs, demonstrating a basic level of understanding of the knowledge and skills associated with AI • Able to provide acceptable solutions to real-world problems • Demonstrate an adequate level of knowledge of AI • Ability to make use of some techniques and knowledge and apply them to familiar situations
Fail (F)	• Achieve none of the six CILOs, with little understanding of the associated concepts and underlying methodologies • Unable to provide solutions to simple problems • Knowledge of AI falling below the basic minimum level • Unable to apply techniques and knowledge to situations or problems

Course Content and CILOs Mapping:

Content		**CILO No.**
I	Introduction to AI	1
II	Basics of Deep Learning	2,5
III	Convolutional Neural Network	1,2,4,5
IV	Recurrent Neural Network	1,2,4,5
V	Attention Mechanism and Transformer Architecture	1,2,4,5
VI	Basics of Generative AI	1,3,4
VII	Foundation Models and Large Language Models	1,3,6
VIII	Multimodal Deep Learning	1,3,4,6

References:

- Stuart Russell and Peter Norvig, Artificial Intelligence, A Modern Approach, Prentice Hall, 4th Edition, 2020.
- Michael Negnevitsky, Artificial Intelligence: A Guide to Intelligent Systems, Addison Wesley, 3rd Edition, 2011.
- Vladimir N. Vapnik, Statistical Learning Theory, Wiley, 1998.
- Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, MIT, 2016.
- Tom M. Mitchell, Machine Learning, McGraw-Hill 1997.
- Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
- Kevin P. Murphy, Machine Learning: A Probabilistic Perspective, MIT, 2012.
- Trevor Hastie, Robert Tibshirani and Jerome Friedman, The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Springer, 2009.
- Aston Zhang, Zachary C. Lipton and Mu Li, Dive into Deep Learning, Cambridge University Press, 2024
- Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, Russ Altman, et al., On the Opportunities and Risks of Foundation Models, ArXiv, 2022.
- Danny Hernandez, Tom Brown, Tom Conerly, Nova DasSarma, et al., Scaling Laws and Interpretability of Learning from Repeated Data, ArXiv, 2022.
- Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, et al., Emergent Abilities of Large Language Models, Transactions on Machine Learning Research (TMLR), 2022.

Course Content:

Topic

I. Introduction to AI
   A. History of AI development
   B. Overview of state-of-the-art AI approaches
   C. Modern AI applications

II. Basics of Deep Learning
   A. Feedforward neural networks basics
   B. Loss minimization, backpropagation, and optimizers
   C. Overfitting, cross validation, and regularization techniques
   D. Application of feedforward neural networks

III. Convolutional Neural Network
   A. Convolutional neural network (CNN)
   B. CNN architectures
   C. Image classification and object detection using CNNs
   D. Video understanding using CNNs

IV. Recurrent Neural Network
   A. Learning word embeddings
   B. Recurrent neural network (RNN)
   C. Variants of RNNs
   D. Sequence-to-sequence modeling

V. Attention Mechanism and Transformer Architecture
   A. Attention mechanism
   B. Transformer Model
   C. Transformer for language modeling
   D. Transformer for vision applications
   E. Pre-trained language models

VI. Basics of Generative AI
   A. Fundamental concepts of generative AI
   B. Generative adversarial network
   C. Denoising diffusion probabilistic models

VII. Foundation Models and Large Language Models
   A. Scaling law and emergent behavior
   B. Post-training and retrieval-augmented generation (RAG)
   C. Prompt engineering
   D. Security, privacy, and ethics for large language models

VIII. Multimodal Deep Learning
   A. Fusion of multiple modalities
   B. Contrastive language-image pre-training (CLIP)
   C. Multimodal foundation models
   D. Visual question answering
   E. Other applications of multimodal learning